Homework

Homework 12 — Projection and intersection — 55 min.

In an orthonormal frame of space, consider the point $M(3,0,0)$ and the planes $\mathcal{P} : x + 2y + 2z = 9$ and $\mathcal{Q} : x - y = 0$.

1. Give the coordinates of a non-zero normal vector to $\mathcal{P}$.
2. Determine the coordinates of the orthogonal projection H of M onto $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$.
3. Show that $\mathcal{P} \cap \mathcal{Q}$ is a line and give a parametric representation of it.
4. Does the point H belong to this line? Justify.
5. Determine the coordinates of the point K of this line whose third coordinate (the "cote") is zero. Calculate MK, then compare MK and MH.